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A Feasibility Study of Video-Based Performance Data Extraction for Football Referee Self-Analysis Applications

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A Feasibility Study of Video-Based Performance Data Extraction for Football Referee Self-Analysis Applications

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Abstract

Football is one of the world's most popular sports, yet the development of performance analysis tools has primarily focused on players and teams, leaving limited support for match officials. For football referees, positioning and proximity to active play are important factors for decision-making. This study addresses this gap through three connected research questions concerning domain understanding, technical feasibility, and the design and evaluation of a referee self-analysis concept based on objective spatial performance metrics from standardized match recordings.

The study combines key performance indicator (KPI) identification, technical development, conceptual application design, and expert evaluation to assess both the technical viability of the proposed system and its potential practical value for referee development. These parts correspond to the three research questions by first identifying relevant referee performance indicators, then assessing whether objective metrics can be extracted from match recordings, and finally exploring how the extracted data can be presented and evaluated in a self-analysis prototype. Based on literature, domain knowledge, and expert input, relevant referee performance indicators were identified and used to guide the development of the Proof of Concept. By implementing computer vision techniques and homography calibration, the system tracks the referee's movement and projects video coordinates onto a two-dimensional representation of the pitch. In addition to the technical pipeline, the study presents a conceptual design for a web-based self-analysis application tailored to referees.

The results indicate that it is technically feasible to generate positioning heatmaps and calculate basic movement metrics using existing camera infrastructure. In the technical analysis, 17 video clips from two different matches were processed. For the primary dataset, consisting of 14 clips, the system calculated a total covered distance of 1,027.7 meters. An isolated five-second sprint interval was also analyzed, in which the system calculated a covered distance of 23.3 meters. Furthermore, the expert evaluation suggests that combining objective spatial data with a user-centered application design has potential to support referee education and self-analysis. The study concludes that accessible video-based spatial analysis can provide a foundation for future tools aimed at improving referee development outside elite-level contexts.

Keywords

Football Referees, Performance Analysis, Computer Vision, Data Pipeline, Proof of Concept, Sports Technology, Self-analysis, Position Tracking, Homography Calibration

Sammanfattning

Fotboll är en av världens mest populära sporter, men utvecklingen av prestationsanalysverktyg har främst fokuserat på spelare och lag, vilket har lämnat ett begränsat stöd för domarteamet. För fotbollsdomare är positionering och närhet till det aktiva spelet viktiga faktorer för beslutsfattande. Denna studie adresserar detta gap genom tre sammanhängande forskningsfrågor som behandlar domänförståelse, teknisk genomförbarhet samt design och utvärdering av ett självanalyskoncept för domare baserat på objektiva rumsliga prestationsmått från standardiserade matchinspelningar.

Studien kombinerar identifiering av nyckelprestationsindikatorer, teknisk utveckling, konceptuell applikationsdesign och expertutvärdering för att bedöma både det föreslagna systemets tekniska genomförbarhet och dess potentiella praktiska värde för domarutveckling. Dessa delar motsvarar de tre forskningsfrågorna genom att först identifiera relevanta prestationsindikatorer för domare, därefter undersöka om objektiva mått kan extraheras från matchinspelningar och slutligen utforska hur den extraherade datan kan presenteras och utvärderas i en självanalysprototyp. Baserat på litteratur, domänkunskap och expertinput identifierades relevanta prestations-indikatorer för domare, vilka användes för att vägleda utvecklingen av Proof of Concept-lösningen. Genom att implementera tekniker inom datorseende och homografikalibrering spårar systemet domarens rörelser och projicerar videokoordinater till en tvådimensionell representation av fotbollsplanen. Utöver den tekniska pipelinen presenterar studien ett expertinformerat konceptuellt förslag på en webbaserad självanalysapplikation anpassad för domare.

Resultaten indikerar att det är tekniskt genomförbart att generera positionerings-heatmaps och beräkna grundläggande rörelsemått med hjälp av befintlig kamerainfrastruktur. I den tekniska analysen bearbetades 17 videoklipp från två olika matcher. För huvuddatasetet, som bestod av 14 klipp, beräknade systemet en total löpdistans på 1027,7 meter. Ett isolerat sprintintervall på fem sekunder analyserades också, där systemet beräknade en tillryggalagd distans på 23,3 meter. Vidare antyder expertutvärderingen att kombinationen av objektiv rumslig data och en användarcentrerad applikationsdesign har potential att stödja domarutbildning och självanalys. Studien drar slutsatsen att tillgänglig videobaserad rumslig analys kan utgöra en grund för framtida verktyg som syftar till att förbättra domarutveckling utanför elitnivå.

Nyckelord

Fotbollsdomare, Prestationsanalys, Datorseende, Datapipeline, Proof of Concept, Sportteknologi, Självanalys, Positionering, Homografikalibrering

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1 Introduction

Football is a highly popular sport globally with around 3.5 billion fans worldwide [1]. In Sweden, there were 440,466 registered players above the age of 12 in 2023, and football accounts for nearly 40% of all sporting activities in the country [2]. Given this immense participation and the cultural significance of the sport, ensuring a consistent and fair experience for players and officials alike is of high importance. While performance analysis and data-driven tools have become increasingly established in football, particularly for analyzing players and teams [3, 4], corresponding tools to support referee development have not seen the same level of technical adoption.

A study observing referees in the Spanish third division by Gomez-Carmona and Pino-Ortega showed that almost every fifth decision regarding technical infractions was incorrect [5]. This can, in some cases, alter the flow of the game, and in the worst case, change the entire outcome. It is therefore critical to keep pace with the development of refereeing by building feedback systems that improve their knowledge and self-awareness.

One important aspect of officiating is the referee's distance to the ongoing play. A study by Johansen and Erikstad concludes that a referee is most likely to make a correct decision when positioned approximately 10 meters away from the active game [6]. Another key factor is the angle to the ball; standard practice in the *Laws of the Game 25/26* from IFAB is for the referee to maintain a diagonal motion across the field to ensure the best possible sightlines [6, 7].

Currently, commercial systems such as Veo Analytics provide in-depth analysis for teams and players, including shot maps, pass strings, and movement radars [4]. While recent updates to these optical tracking systems include the capability to detect and display the referee on a 2D radar map, the advanced analytical features remain exclusively focused on player optimization [8]. The referee is visually tracked as a coordinate point, but no objective spatial statistics, such as distance covered or positioning heatmaps, are extracted to support officiating evaluation. There is currently no integrated solution that utilizes this available match footage to provide the same level of objective feedback to the referees.

This untapped potential of existing match recordings is particularly relevant within the current infrastructure of Swedish football. Since 2022, the Swedish Football Association has maintained a strategic partnership with Veo, enabling the widespread livestreaming and recording of matches across various non-elite divisions [9, 10]. Utilizing this already established infrastructure presents a significant opportunity to democratize access to objective performance evaluation for referees across all levels.

To address the lack of technical support for match officials highlighted above, this study adopts a practical approach grounded in the author's extensive domain expertise, including eight years as an active football referee and over three years as a referee coach. By combining these first-hand insights with theoretical literature and expert validation, the study systematically identifies the core analytical needs of grassroots referees. These identified needs then form the foundation for developing a targeted technical Proof of Concept, exploring how standard match video can be leveraged to establish an accessible framework for independent referee self-analysis.

1.1 Problem

The central problem addressed in this project is the lack of objective, accessible tools for referee performance evaluation. While players and teams benefit from advanced video analysis and GPS tracking to monitor their performance, referees primarily rely on subjective feedback from observers [11, 3]. At the elite level, referees have access to some analytical resources and regular feedback. However, for the vast majority of referees at lower or semi-professional levels, systematic feedback is more infrequent or nonexistent.

Currently, there is no accessible tool that allows these referees to independently upload their match videos, view objective statistics regarding their positioning, and perform meaningful self-analysis. Providing an application where referees can easily input their match footage to extract performance metrics (such as distance to the ball and movement patterns) could significantly aid their professional development and improve the accuracy of their decision-making.

Therefore, the project is structured around three connected research questions addressing domain understanding, technical feasibility, and design and evaluation.

1.2 Purpose

A referee's performance can be evaluated from several different perspectives. Based on the literature review and domain knowledge, several areas appear particularly relevant for referee self-analysis: positioning on the pitch, movement patterns, physical workload, and distance to active play. These areas can be translated into practical key performance indicators (KPIs), such as positioning patterns, distance covered, sprint intervals, and proximity to play. They are relevant because they may influence the referee's ability to maintain an appropriate position, follow the diagonal system of control, and make decisions from a suitable distance and angle.

To address the lack of objective evaluation tools for referees at non-elite levels, this project investigates whether these indicators can be extracted from standard match video and translated into a practical self-analysis concept. The purpose is not to deliver a fully validated commercial tool, but to examine whether a Proof of Concept can generate objective referee performance data from standard Veo match recordings.

By investigating this feasibility, the study aims to contribute to bridging the technological gap between players and referees by demonstrating whether existing AI and computer vision techniques used for player tracking can be adapted for referee evaluation. The results are intended to be used to design conceptual mockups of a self-analysis application and to gather qualitative feedback from referee developers. This will illustrate how such a tool could function in practice and whether the concept is of practical interest for supporting referee self-analysis and development at different competitive levels.

Based on this purpose, the study is guided by three connected research questions. The first question focuses on the domain understanding needed to identify relevant referee performance indicators. The second question addresses the technical feasibility of extracting objective referee metrics from standard football match recordings. The third question focuses on how these metrics can be visualized in a self-analysis application and evaluated through expert feedback.

RQ1 – Domain understanding: What information and performance indicators do referees and referee coaches consider valuable for referee performance analysis?

RQ2 – Technical feasibility: To what extent can objective referee performance metrics be extracted from standard football match recordings using computer vision techniques?

RQ3 – Design and evaluation: How can extracted referee performance metrics be visualized and presented in a self-analysis application, and what insights do expert referee coaches gain from interacting with such a prototype?

1.3 Goal

The goal of this project is to establish a functional Proof of Concept for extracting objective referee performance data from video, and to conceptualize how this data can be utilized in a self-analysis application. This has been divided into the following sub-goals:

1. **KPI and Evaluation Criteria Identification:** Identify relevant referee performance indicators based on literature and domain experience. These indicators form the basis for both the technical development and the later evaluation of the proposed self-analysis concept.

2. **Data Extraction and Processing:** Develop a computer vision pipeline that tracks the referee in standard match videos using a semi-automatic approach, applies homography to translate pixel coordinates into physical X/Y coordinates on a 2D pitch, and processes the extracted coordinates to calculate objective performance metrics such as total distance covered, positioning density, and sprint-related metrics.

3. **Visual Representation and Self-Analysis:** Design conceptual mockups of a user interface that displays and visualizes the extracted performance indicators, for example through heatmaps and dashboards, to support objective referee self-analysis.

4. **Evaluation and Iterative Feedback:** Gather qualitative feedback from a referee developer to evaluate the practical relevance of the identified indicators, the interface concept, and the overall self-analysis workflow. The feedback is then used to refine the prototype and identify future development needs.

1.4 Research Methodology

This project employs a mixed methods approach, combining both quantitative and qualitative methods to address the research problem. The quantitative research focuses on the technical development through an experimental setup to build and evaluate the feasibility of a Proof of Concept video analysis pipeline. This involves applying computer vision techniques, specifically the CSRT object tracking algorithm and homography transformations, to extract empirical data such as physical coordinates and movement metrics from standard match video recordings. The reliability and plausibility of the pipeline are assessed by analyzing its ability to generate consistent positioning data across various match situations.

The qualitative research is utilized for the conceptual design and evaluation phases. This involves developing user interface mockups for a proposed referee self-analysis application. Subsequently, qualitative data was gathered through an in-depth interview and feedback session with a referee developer at the Stockholm Football Association to assess the perceived value and practical utility of the conceptualized application.

The conceptual design process was also informed by participatory and user-centered design principles, since the needs identification and prototype evaluation involved domain expertise from referee development. In this project, this was applied in a limited form through expert input rather than through a full participatory design process with multiple end-users [12].

1.5 Delimitations

To ensure the project remains focused and achievable within the given timeframe, several delimitations have been established regarding the data source, the evaluation scope, and the final application delivery.

First, regarding the video source, the data extraction pipeline is specifically developed and tested using match footage recorded by Veo cameras. This limitation was chosen because Veo is a standard recording tool utilized by the Swedish Football Association for non-elite leagues, making the extracted data highly relevant for the target demographic.

Second, the system is strictly limited to extracting objective spatial data, primarily the referee's positioning and movement patterns. It does not attempt to automatically evaluate, grade, or judge the correctness of the referee's actual decisions on the pitch, such as fouls or offsides. Instead, the

system is limited to spatial and physical performance indicators, such as positioning, movement patterns, and proximity to active play, which may influence the referee's opportunity to make well-positioned decisions.

Finally, regarding the application scope, the project will not deliver a fully programmed, functional mobile or web application. Instead, the output concerning the user interface is restricted to conceptual high-fidelity mockups developed in Figma. This delimitation allows the research to focus heavily on solving the core technical challenges of the data pipeline while still demonstrating the practical end-use through conceptual design and qualitative feedback.

1.6 Structure of the thesis

Chapter 2 presents the foundational background necessary to understand the project, including the development of sports analytics, current referee evaluation methods, relevant computer vision techniques, technical frameworks, validation approaches, and related work. Chapter 3 details the research methodology, describing the mixed-methods approach, the needs identification process, the technical development process, the conceptual design process, and the expert evaluation. Chapter 4 presents the results of the feasibility study, including the identified needs and KPI mapping, the technical prototype development, the technical extraction results, the conceptual application prototype, the qualitative expert evaluation, and the post-evaluation design refinements. Chapter 5 analyzes and discusses the reliability, validity, and practical implications of the system. Chapter 6 concludes the degree project, summarizes the main findings, discusses limitations, and presents suggestions for future work.

2 Background

This chapter provides the theoretical and domain-specific background needed to understand the project. It introduces the development of sports analytics in football, current approaches to referee evaluation, and the computer vision techniques that form the technical basis of the proposed solution. The chapter also reviews previous research and existing systems related to player tracking, referee performance analysis, and video-based feedback. By comparing these approaches, the chapter identifies the gap addressed in this thesis: the limited availability of accessible, objective analysis tools for referee development at non-elite levels.

2.1 The Evolution of Sports Analytics in Football

The sports industry has undergone a significant transformation over the past two decades, shifting from observational evaluations to more objective, data-driven analysis. In association football, the integration of advanced data analytics is now considered a fundamental component for evaluating player performance and optimizing team tactics [3]. Historically, performance analysis in football relied heavily on manual notation methods, which were inherently limited by human error, processing time, and a lack of continuous movement data [13]. Today, Electronic Performance and Tracking Systems (EPTS) have become increasingly prominent. These officially recognized systems are used to automatically capture vast amounts of spatio-temporal data during both training and competition [14].

The development of EPTS technologies is generally categorized into wearable devices and camera-based optical tracking systems [15]. Wearable technology typically utilizes Global Navigation Satellite Systems (GNSS) or Local Positioning Systems (LPS) to track player movement [14]. These physical sensors are attached to the players to provide instantaneous physical metrics, such as total distance covered, high-speed running, and acceleration profiles [15]. While highly accurate for monitoring individual physical loads, wearable devices require the physical equipping of every participant and inherently lack the spatial context of the ball and the overall pitch layout [15].

To address the contextual limitations of physical sensors, there has been a widespread adoption of camera-based optical tracking systems [15]. Modern optical tracking leverages computer vision and machine learning algorithms to detect and track multiple objects, including players, referees, and the ball, simultaneously from video footage [8, 16]. These systems function by applying spatial transformations to translate video pixels into physical X/Y coordinates on a 2D top-down representation of the pitch [16].

This coordinate projection enables the generation of advanced visual analytics, such as movement trajectories, spatial radars, and density heatmaps [3]. Commercial solutions, such as Veo Analytics, have further democratized this technology by providing automated, AI-driven tracking without the need for manual camera operators, making complex spatial data accessible for grassroots and semi-professional levels [4]. These systems successfully detect both players and referees, projecting them onto a 2D spatial radar, see Figure 1 [8]. However, while the optical tracking is capable of capturing all moving actors on the pitch, both existing research and commercial applications remain largely focused on player and team optimization. The referee is detected by the system but is frequently overlooked in the application of spatial evaluation models [3]. Finally, advanced data analytics in football functions as a complementary tool to support, rather than replace, professional human expertise [17].

The accessibility of such technologies is also relevant from a sustainability perspective, since sport has been recognized as an enabler of sustainable development and social inclusion [18]. In this

context, developing lower-cost and more accessible analysis tools may contribute to reducing differences in development opportunities between elite and non-elite football environments.



Figure 1: **Example of a 2D radar map generated by Veo Analytics, showing the projected positions of players and the referee [19].**

While the technical infrastructure for tracking exists, the practical application for refereeing requires a specific understanding of the official's role. A referee is the sole authority on the pitch, tasked with maintaining the laws of the game while constantly adjusting their position to ensure optimal sightlines without interfering with the flow of play [7]. This creates a unique spatial challenge where the official must balance proximity to the ball with a broad view of the surrounding players, a task that has historically been evaluated through subjective human observation [6].

Sport has been recognized as an important enabler of sustainable development, particularly through its contribution to tolerance, respect, health, education, and social inclusion objectives [18]. In addition, European sport policy emphasizes that sport should be fair, ethical, safe, and accessible, with fair play and respect as central principles [20]. In football, the referee has full authority to enforce the Laws of the Game and to control the match in cooperation with the other match officials [7]. Therefore, supporting referee development can also be viewed as part of strengthening fair and well-functioning football environments, particularly when accessible feedback tools can be made available beyond elite-level contexts.

2.2 Referee Evaluation and Performance Metrics

At the majority of competitive levels, the evaluation of football referees currently relies almost exclusively on human observation and qualitative reporting frameworks [11]. Official referee observers typically use standardized evaluation matrices or rubrics to visually assess a referee's performance across various categories, such as physical endurance, body language, and spatial positioning [11]. For example, within Swedish association football, the Stockholm Football

Association utilizes a specific evaluation template for referee observers consisting of 16 distinct performance points [21]. However, this standard observational method is inherently subjective, as it relies on the evaluator's visual estimation from the stands without the support of continuous, objective spatial tracking data [5].

To move toward a more quantifiable evaluation of referee performance, sports science has identified specific spatial metrics that correlate with accurate decision-making [5]. A primary metric in this context is the physical distance between the referee and the foul or incident [6]. Empirical studies demonstrate that a referee's ability to make correct decisions, particularly regarding technical infractions and penalty area incidents, is significantly optimized when they are positioned at a distance of approximately 10 to 15 meters from the play [6].

Furthermore, effective officiating is highly dependent on the viewing angle and an unobstructed field of vision [5]. To standardize optimal positioning, the International Football Association Board (IFAB) dictates the "Diagonal System of Control" (DSC) as the fundamental movement protocol for referees [7]. This system requires the referee to patrol a flexible diagonal path across the pitch, typically extending from one penalty area to the opposite, while the assistant referees cover the remaining touchlines [7]. The primary objective of the DSC is to ensure that the match officials maintain a "pincer movement," keeping the ball and active players between them to provide multiple viewing angles and avoid obstructed lines of sight [7]. Despite these clear spatial guidelines, human observers are incapable of manually calculating exact distances or validating the exact diagonal trajectory over a 90-minute match, highlighting a critical need for automated spatial tracking tools [11, 5].

The shift from external observation to objective self-analysis is supported by pedagogical theories on expert learning. Research indicates that video-based self-reflection allows referees to bridge the gap between their subjective perception on the pitch and the objective reality of the match [22]. By systematically reviewing their own decision-making through video footage, officials can identify specific cognitive biases and improve their situational awareness [22]. This process of reflective practice has been shown to significantly enhance decision-making accuracy, as it enables referees to internalize optimal positioning patterns and learn from previous errors in a controlled, off-field environment [22]. As domain input for this project, Suhel Youssef, a referee developer at the Stockholm Football Association and the industrial adviser for this project, emphasized that referees can learn a great deal from conducting self-analysis. He explained that referees learn by watching themselves and reviewing their own performance, both in terms of what they do well and what they can develop. For example, a referee may perceive that they have good movement patterns and that they move out toward the sidelines effectively, but by watching video footage they can gain a better understanding of how this can be improved.

In summary, the literature reviewed in this section indicates that referee performance analysis should consider spatial positioning, proximity to active play, viewing angle, diagonal movement behavior, physical workload, and structured video-based self-reflection. These aspects therefore provide the basis for the KPI identification described in the methodology chapter.

2.3 Computer Vision in Sports

Computer vision is an interdisciplinary field of artificial intelligence that focuses on enabling computers to interpret and understand the visual world from digital images and videos [23]. This technology is a cornerstone in many advanced industries; for example, it is used in the medical field to aid in diagnosing conditions from X-rays and in the automotive industry to enable autonomous vehicles to identify pedestrians and obstacles [23]. In the context of athletic performance, computer

vision transforms raw video frames into quantitative data by identifying and locating specific objects, such as players or a ball, in every individual frame [16].

In this thesis, these techniques are relevant because the proposed system must first identify the referee in the video, then follow the referee across frames, and finally translate the tracked video position into physical pitch coordinates. The following section therefore introduces the three main technical steps required for the system: detection, tracking, and coordinate projection.

To overcome the limitations of manual observation in sports, modern analytics leverages computer vision to automatically extract continuous spatial data from standard video footage [16]. Rather than relying on physical wearable sensors, optical tracking systems use advanced machine learning algorithms to interpret the visual information captured by cameras, enabling an objective analysis of spatial metrics [16].

The foundational step in optical tracking is object detection. State-of-the-art sports tracking systems frequently utilize Convolutional Neural Networks (CNNs), such as the You Only Look Once (YOLO) architecture, due to its optimal balance of high detection accuracy and real-time processing capabilities [24]. In the context of football, these detection models are trained to identify key actors on the pitch, primarily the players and the ball, and enclose them within bounding boxes in every individual video frame [16, 24].

However, object detection alone only provides isolated spatial data per frame. To analyze continuous movement, such as player runs or tactical formations, the detection model must be paired with object tracking algorithms [25]. Algorithms such as DeepSORT (Simple Online and Realtime Tracking with a Deep Association Metric) associate the detected bounding boxes across sequential frames [25]. By extracting both the motion kinematics and the visual appearance of the actors, the system assigns a unique identification (ID) to each tracked object and maintains that ID throughout the match, generating a continuous trajectory [25].

Finally, to calculate physical distance, such as the total distance covered by a player or the proximity between an attacker and a defender, the video data must be translated into real-world measurements. This is achieved through a perspective transformation, typically utilizing a homography matrix [26]. By mapping the video pixels to a 2D top-down representation of the football pitch, the system converts the pixel coordinates of the bounding boxes into physical X and Y coordinates measured in meters [26, 27].

Commercial systems, such as Veo Analytics, utilize this coordinate projection to generate real-time 2D radar maps for tactical player evaluation [4]. While existing commercial applications primarily apply this pipeline to optimize player performance, this exact technological framework of detection, tracking, and coordinate projection provides the fundamental foundation required to objectively measure referee positioning and address the limitations of human evaluation [16, 28].

2.4 Technical Framework

This section outlines the theoretical background of the technical frameworks and software tools relevant to the development of video analysis systems and conceptual user interfaces.

2.4.1 Programming Languages and Data Processing

Python is a high-level programming language that has established itself as an industry standard in data science, machine learning, and computer vision, largely due to its extensive ecosystem of open-source libraries [29]. A prominent alternative for system development is C++, which offers superior execution speed and detailed memory control, qualities often preferred in real-time production environments [30]. However, developing in C++ generally involves a steeper learning curve and

longer development times for prototyping compared to Python [31]. MATLAB is another frequently utilized tool in academic research for advanced mathematical modeling, but its applicability in open and distributed web architectures is constrained by its proprietary, licensed nature [32].

For handling the spatiotemporal data derived from automatic video tracking, data processing libraries such as Pandas are commonly used. Raw tracking data often contains high-frequency noise and minor positioning fluctuations, referred to as bounding box jitter or measurement noise [33]. To convert these raw coordinate steps into realistic physical metrics, digital signal smoothing techniques are applied. A standard statistical method for reducing variance in time-series datasets is the rolling window moving average, where each data point is replaced by the mean of a specified number of surrounding frames [34]. This filters out sudden, unrealistic spikes in velocity without destroying the underlying macro-trends of the movement.

2.4.2 Computer Vision and Object Tracking

Open Source Computer Vision Library (OpenCV) is one of the most widely adopted frameworks for computer vision, featuring over 2,500 optimized algorithms for tasks ranging from motion tracking to camera calibration [35]. In sports analytics, OpenCV is frequently used for perspective transformation, or homography. Homography is a mathematical mapping based on a projective transformation between two planes, and in football video analysis it can be used to map image coordinates from a camera view onto a standardized top-down representation of the pitch using known field reference points [26, 27].

For tracking objects across consecutive video frames, algorithms based on Discriminative Correlation Filters (DCF) are commonly employed. These filters learn a template to distinguish between the target object and its surrounding background [36]. Within this category, the Channel and Spatial Reliability Tracker (CSRT) introduces spatial and channel reliability weights. This allows the tracker to dynamically adapt to changes in the target's shape and handle partial occlusions, enhancing tracking stability at the expense of higher computational complexity compared to simpler models like the Kernelized Correlation Filter (KCF) [37]. As an alternative to these traditional algorithmic methods, modern deep learning models, such as YOLO (You Only Look Once) architectures, are increasingly used. While these neural networks can achieve significantly higher accuracy in complex scenarios with multiple overlapping objects, they require massive annotated datasets and substantial GPU computational power for both training and inference [24, 38].

2.4.3 UI/UX Design and Prototyping

Figma is a cloud-based design platform that enables collaborative development of user interfaces and interactive prototypes directly within a web browser, eliminating the need for local software installations [39]. Figma also includes AI-assisted design features that can support the generation of initial interface layouts and design components [39]. Alternative software for interface design includes Sketch and Adobe XD. Sketch provides design and prototyping functionality but requires a Mac running macOS, which limits cross-platform development [40]. Adobe XD also supports interactive design, prototyping, and sharing of user interfaces [41]. Compared with alternatives such as Sketch and Adobe XD, Figma offers browser-based interface design, real-time collaboration, and interactive prototyping, making it suitable for rapid development of conceptual user interfaces [39].

2.5 Validation of Sports Technology Systems

The development of sports technology systems requires both technical and practical forms of validation. Technical validation concerns whether the system produces accurate and reliable data, while practical validation concerns whether the generated information is understandable, relevant, and useful for the intended users [42]. In sports analytics, this distinction is important because technically measurable outputs only become valuable if they can support meaningful performance development in practice [42].

2.5.1 Technical Validation of Spatiotemporal Data

In sports analytics, technical validation of tracking systems commonly involves assessing the accuracy of extracted spatiotemporal variables, such as position, distance covered, and speed [15]. For player tracking systems, this is often done by comparing video-derived or optical tracking data against a reference system, such as wearable GNSS, local positioning systems, or other validated EPTS technologies [15]. In video-based systems that rely on perspective transformation, calibration accuracy is also important, since homography-based methods depend on known reference points in the image and on the physical playing surface [26, 27]. When external ground-truth data is unavailable, visual inspection and plausibility checks may provide an initial indication of whether the extracted movement patterns are reasonable, but they should not be treated as a substitute for formal validation against an independent reference system [15].

2.5.2 Practical Validation and User Relevance

In addition to technical accuracy, sports technology systems must also be evaluated in relation to their practical usefulness [42]. A system intended for performance development needs to present information in a way that domain users can interpret and apply [42]. Qualitative semi-structured interviews can be used to gather in-depth insights into participants' experiences, perspectives, and evaluations of a given topic [43]. This is particularly relevant for conceptual interfaces, where the goal is not only to test whether the system can calculate specific metrics, but also whether these metrics are meaningful for the intended learning or development process [42].

2.6 Related Work

The application of technology in football analytics has been extensively researched, yet the vast majority of this work remains disproportionately focused on player performance [44]. Player-focused football analytics commonly extract metrics such as player trajectories, physical movement variables, tactical formations, and ball-related events, demonstrating that the underlying technology is well established for player analysis [3, 44]. Existing computer vision research frequently utilizes algorithms like YOLO to extract player kinematics, tactical formations, and ball trajectories, while the referee is either ignored entirely or simply detected without further spatial analysis [44].

When related work has focused on referee evaluation, it has almost exclusively been conducted at the highest professional elite levels [45, 46]. Researchers have predominantly analyzed referee performance by quantifying physical workloads, such as total distance covered, high-speed running, and heart rate metrics, using wearable Global Positioning System (GPS) technology or Local Positioning Systems (LPS) [45]. While highly accurate, these sensor-based approaches require the physical equipping of the referee, which is logistically complex and financially unfeasible for lower competitive levels.

Furthermore, in the limited instances where optical tracking (spatio-temporal data) has been used to analyze a referee's tactical positioning, such as measuring their proximity to the ball or adherence to the diagonal movement, researchers have relied on highly expensive, multi-camera tracking systems permanently installed in professional stadiums [46].

According to Suhel Youssef, the Swedish Football Association currently utilizes a simplified internal system for match review. This platform allows users to upload match recordings, mark specific timestamps for video clips, and add qualitative comments. Additionally, the system displays reported match events such as goals and disciplinary sanctions. While this provides a functional baseline for video review, the tool is primarily restricted to internal use within the association's elite development programs and lacks integrated objective spatial data or automated performance metrics.

Consequently, a significant research gap exists regarding the democratization of referee evaluation. Currently, there is a lack of literature addressing how single-camera, commercially available optical tracking systems (which are increasingly common at the grassroots and semi-professional levels) can be utilized to automatically extract spatial performance metrics for referees [44, 46]. This thesis aims to bridge this gap by transitioning automated spatial referee evaluation, such as analyzing referee proximity to play and adherence to the Diagonal System of Control, from elite, multi-camera stadium setups to accessible, AI-driven single-camera frameworks.

3 Methods

This chapter describes the research process and the methods used to develop and evaluate the proposed system. The project combines quantitative technical development with qualitative expert evaluation in order to assess both the feasibility of the video analysis pipeline and the practical relevance of the application concept.

3.1 Research Process and Preliminary Validation

The project was structured as a sequential feasibility study consisting of three main phases. The first phase focused on initial needs identification and domain validation. The second phase involved developing the technical Proof of Concept, which included building a technical foundation to retrieve coordinate data and conceptualizing a user interface. The third phase consisted of an expert evaluation to facilitate refinement based on professional feedback. The sequential flow of this research process is illustrated in Figure 2.



Figure 2: Overview of the sequential research methodology

3.2 Needs identification

To ensure that the developed Proof of Concept addresses genuine, practical challenges in football refereeing, the first phase focused on rigorous needs identification. This phase relied heavily on personal domain experience as a former referee and referee coach, supplemented by theoretical insights gathered from the literature review. By combining these sources, a specific set of core analytical needs for non-elite referees was identified and detailed. To ensure alignment with actual referee development, these preliminary needs were reviewed and confirmed by Suhel Youssef, a referee developer at the Stockholm Football Association, before the technical development commenced. The identified needs and KPIs were then summarized in a requirements mapping matrix, which is presented in the results chapter and used to connect the domain needs to the technical and conceptual components of the proposed system.

3.3 Development of the Data Extraction Pipeline

Based on the limitations identified in the background chapter, the technical development focused on assessing whether accessible single-camera footage could be used to extract objective spatial metrics for referee evaluation. In this thesis, spatial performance metrics refer to measurable outputs that describe where the referee is positioned on the pitch, how the referee moves over time, and how this positioning relates to relevant match situations. The main metrics considered in the project were:

- **Total distance covered**, referring to the accumulated distance travelled by the referee during the analysed video clips.
- **Sprint intervals**, referring to selected time periods where distance, duration, and speed can be analysed.
- **Positioning heatmaps**, referring to visual representations of where on the pitch the referee spent the most time.
- **Distance to active play (the ball)**, referring to the referee's proximity to the active game situation over time.

- **Positioning during disciplinary incidents**, referring to the referee's location on the pitch when yellow or red cards are issued.

Together, these metrics were selected because they can provide visual and quantitative feedback on the referee's movement patterns, physical workload, proximity to play, and positioning during important match events. These intended metrics motivated the use of a customized data pipeline combining object tracking, pitch calibration, and coordinate transformation.

Based on the technical alternatives outlined in Section 2.4, Python and OpenCV were selected for the technical Proof of Concept because they support rapid prototyping and provide established tools for video processing, object tracking, and homography-based coordinate transformation. A semi-automatic tracking approach was used instead of a fully trained deep learning detector, since the project aimed to evaluate feasibility within a limited timeframe without requiring a large, annotated referee dataset.

The pipeline was developed to process selected clips from Veo match recordings and to convert the referee's tracked image position into pitch-based coordinates. The main methodological steps consisted of manual referee initialization, object tracking, pitch calibration, coordinate transformation, and subsequent processing of the extracted coordinates into movement metrics and visualizations. The technical implementation details, including the calibration procedure, tracking interface, smoothing rules, and filtering thresholds, are presented in Section 4.2 as part of the technical prototype development. This methodological setup allowed the project to assess whether standard match footage could provide a sufficient technical basis for extracting referee performance metrics.

3.4 Conceptual Application Design in Figma

With the technical metrics identified, a conceptual design phase was initiated using Figma, a collaborative interface design and prototyping tool [39]. Figma was selected for this project because it enabled rapid development of an interactive, web-based prototype that could be presented and discussed during the expert evaluation. The goal was to visualize the extracted data in a way that supports a referee's self-analysis. This phase involved creating a high-fidelity design proposal that integrated dashboards for heatmaps and physical statistics. The prototyping features in Figma and integrated AI tools were utilized to iterate on interface layouts ensuring the complex spatial data was presented through intuitive components. This design was largely completed before the formal evaluation to provide a concrete artifact for expert review.

3.5 Expert Evaluation and Interview

The final qualitative evaluation was conducted through a 30-minute digital interview and feedback session with Suhel Youssef, the primary domain expert for this project. Since the purpose of the evaluation was to assess the practical relevance of an early-stage Proof of Concept rather than to produce generalizable user-study results, a single domain expert was considered appropriate for this phase. The primary objective of this phase was twofold: first, to have the expert validate that the core domain needs were correctly identified in Phase 1; and second, to assess whether the conceptual UX design (Figma prototype) was intuitive, comprehensible, and practical for an independent external user.

During the session, the overall project aim, the extracted spatial data, and the technical Proof of Concept were presented, followed by a demonstration of the application prototype. The interview followed a semi-structured format. While some topics were discussed continuously during the presentation, the expert was also asked open-ended questions regarding the perceived usefulness of the existing features, whether the solution met the actual needs of referee development, and what additional functionality could be valuable. This key informant approach provided specialized

professional insights into how the system could be used in practice and how the application concept could be further refined.

3.6 Iterative Design Refinement

The final phase of the methodology involved a limited refinement of the application proposal based on the feedback obtained during the expert interview. Due to the project timeframe, only adjustments that could be implemented within the existing Figma prototype were carried out. More extensive suggestions were documented as potential future work. This step ensured that the expert feedback was incorporated into the design process.

4 Results

This chapter presents the results of the feasibility study. The results are organized according to the main phases of the project. First, the identified referee development needs and KPIs are summarized in a requirements mapping matrix. Second, the technical prototype development is presented. Third, the technical extraction results are reported. Fourth, the conceptual application prototype is described. Finally, the qualitative expert evaluation and post-evaluation design refinements are presented.

4.1 Needs Identification and KPI Mapping

The first result of the project was the identification of core referee development needs and their translation into practical key performance indicators (KPIs) and system requirements. As described in Section 3.2, these needs were identified based on the literature review, domain knowledge, and expert input. Four central needs were identified: objective positioning and movement analysis, physical work rate measurement, distance to active play, and an accessible self-analysis workflow. Table 1 summarizes how each need and KPI area was mapped to a corresponding technical or conceptual component of the proposed system, and shows how the selected indicators guided the development of the Proof of Concept.

Table 1: Requirements mapping matrix connecting the identified domain needs from Phase 1 to the technical implementation and conceptualization developed in Phase 2.

Phase 1: Identified Need / KPI Area	Phase 2: Technical PoC Implementation & Conceptualization	Rationale / Objective
Objective Positioning & Movement Analysis: The need to map the referee's overall movement patterns (e.g., to identify overly central positioning or lack of diagonal movement).	**2D Pitch Projection & Heatmap Generation:** Utilizing homography calibration to translate video pixels into 2D coordinates, outputting visual heatmaps.	To provide visual feedback on the referee's coverage of the field.
Physical Work Rate Measurement: The need to measure the referee's physical output and intensity during specific phases or events of the game.	**Spatiotemporal Tracking (Distance & Sprints):** Calculating total running distance and analyzing specific sprints (distance and speed) based on user-defined time intervals.	To objectively quantify physical exertion and provide detailed insights into specific high-intensity moments during the match.
Distance to Active Play: The need to monitor the referee's proximity to the ball throughout the match to evaluate positioning during key decisions and identify potential fatigue.	**Conceptual Graph Design (Figma):** Modeling a "Distance to Ball" time-series graph to illustrate how referee coordinates could be combined with ball-position data in a future system.	To provide a visual analysis of the referee's positioning relative to active play over time.
Accessible Self-Analysis Workflow: The need for a streamlined, intuitive interface	**Conceptual UI/UX Design (Figma):** Development of a high-fidelity interactive	To bridge the gap between complex computer vision data and practical pedagogical

where referees can review objective statistics and reflect on their performance.	prototype mapping out the interface and user journey for independent post-match reflection.	utility, demonstrating how spatiotemporal metrics can be visualized in an actionable dashboard.

4.2 Technical Prototype Development

The system was implemented as a sequential data pipeline consisting of three main stages: pitch calibration, object tracking, and data processing and visualization. The core design decision was to break the problem into modular steps to allow for batch processing of video clips. Figure 3 illustrates this sequential workflow, demonstrating how data from individual clips is processed iteratively before being aggregated into final heatmaps and metrics.

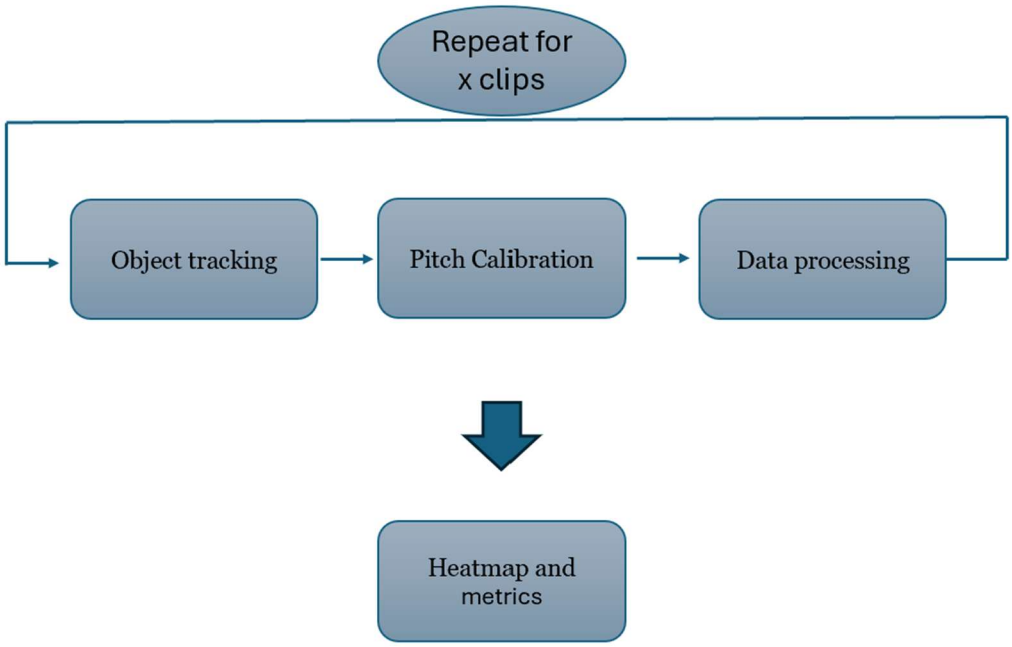


Figure 3: **Flowchart of the data pipeline, showing the iterative processing of individual video clips and the final generation of heatmaps and metrics.**

4.2.1 Pitch Calibration and Homography Calculation

Before extracting movement data, the system must establish the mathematical relationship between the camera's perspective and the physical pitch. A custom Graphical User Interface (GUI) allowed the operator to identify key pitch landmarks from selected video frames and enter the pitch dimensions. If the exact pitch dimensions were unknown, the system could instead use FIFA's 105 × 68 meter standard dimensions as a default reference. These dimensions were saved and reused across clips from the same match, while the visible pitch landmarks were manually selected for each calibration keyframe. In practice, this meant that the operator selected visible pitch landmarks at several frames within each clip, for example at regular intervals such as frames 0, 100, 200, 300, 400, 500, and 600, depending on the clip length and camera movement. Using OpenCV's cv2.findHomography function, a 3 × 3 transformation matrix was calculated for each keyframe. To account for minor camera movements, the system supported multiple calibration keyframes and

interpolated between keyframe-based coordinate estimates when converting tracked pixel positions into physical pitch coordinates. For visual validation, the inverse transformation was used to project the 2D pitch model back onto the video frame; if the projected lines aligned with the visible pitch markings, the calibration was considered plausible, as demonstrated in Figure 4.



Figure 4: **Visual validation of the homography calibration, showing the 2D pitch model projected back onto the video frame.**

4.2.2 Implementation of Object Tracking

With the spatial foundation established, the system utilizes the CSRT algorithm to track the referee's movement frame by frame. The process begins with a manual initialization where the operator defines a bounding box around the referee in the initial frame. The tracked referee position was represented by the bottom-center point of the bounding box, corresponding approximately to the referee's foot position on the pitch.

To ensure data integrity, a semi-automatic "rescue" feature was implemented to handle tracking failures caused by occlusions or rapid movements. If the tracker loses the target, the operator can pause the system, re-initialize the bounding box as illustrated in Figure 5, and resume. During the processed clips, manual re-initialization was only required occasionally, and instances where the referee moved completely outside the broadcast frame were relatively uncommon. A key technical detail is the compensation for human reaction time: when marking the target as "missing", the system retroactively removes the preceding 15 frames of data. This ensures that erroneous points, which are captured in the split second before the operator can react, are excluded from the final dataset. The extracted x and y coordinates are then continuously saved to a CSV file for further analysis.

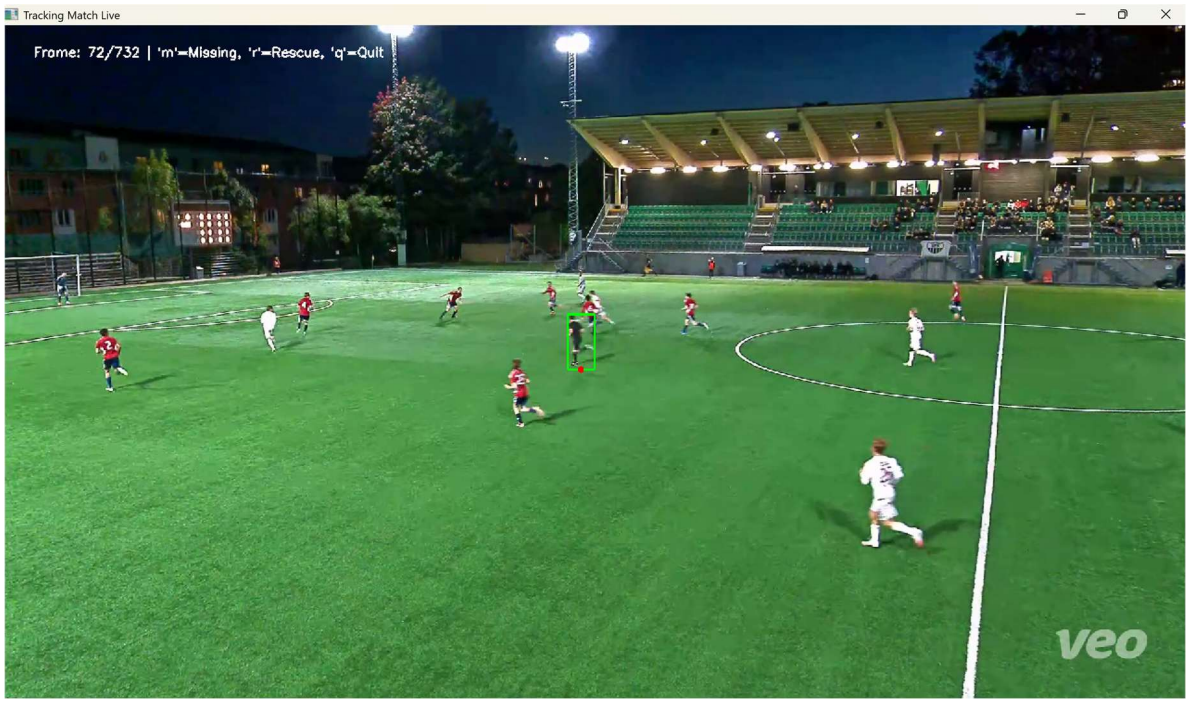


Figure 5: **The CSRT tracking interface during live processing, showing the referee enclosed in a bounding box and the on-screen controls used to mark tracking loss ("missing") and re-initialize the target ("rescue").**

4.2.3 Data Processing and Visualization

The next phase of the pipeline combines the saved pixel tracking data with the calibration keyframes to generate objective metrics and visualizations. The tracked pixel coordinates are transformed into physical pitch coordinates using the homography matrices calculated during calibration. When multiple calibration keyframes are available, the system interpolates between the nearest keyframe-based coordinate estimates to account for minor camera movement.

To account for minor jitter from the tracking bounding box, smoothing was applied before visualization and metric calculation. The movement trajectory used for clip visualization was smoothed using a rolling mean, while the distance and sprint calculations applied an additional 15-frame rolling average to reduce short-term coordinate noise. A physiological filter was then introduced to ensure realistic results. This included a dead zone where movement below 2.0 km/h was registered as zero to remove stationary noise, and a speed cap of 32.0 km/h to limit unrealistic tracking spikes. The Pythagorean theorem was then used to calculate the physical distance between each consecutive smoothed point. The total distance covered was then calculated by summing these stepwise distances across each processed clip. To enable targeted analysis, a manual interval function was also developed. This allows the operator to input specific start and end timestamps for an event, such as a sprint, to calculate duration, max speed, and distance for that specific moment. Finally, a custom Matplotlib function rendered the trajectory onto a scaled 2D pitch model, as shown in Figure 6, and a KDE-based density heatmap was generated to visualize the zones where the referee spent the most time.

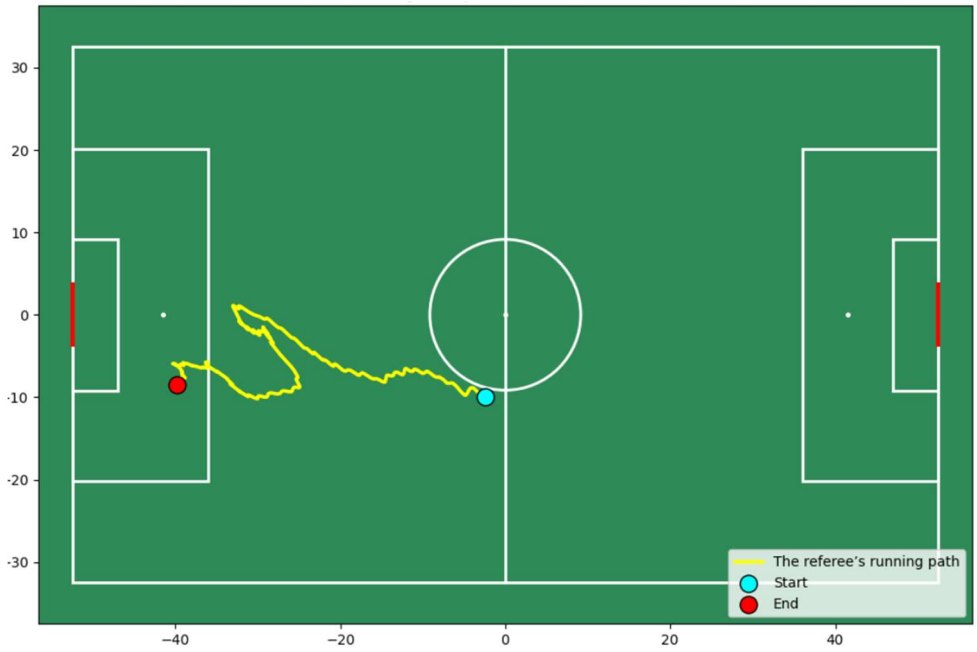


Figure 6: **Visualization of a processed clip using Matplotlib, mapping the referee's physical movement trajectory over a standardized 2D pitch model during one clip.**

4.3 Technical Extraction Results

The primary technical result is the translation of raw video footage into objective spatial data and physical metrics. The resulting spatial outputs and movement metrics were generated from 17 pre-selected video clips across two separate football matches, providing an initial indication of the pipeline's adaptability to different recording environments.

4.3.1 Spatial Visualization

To evaluate the system's ability to map movement patterns, a primary dataset (Match A) consisting of 14 video clips was processed. The aggregated spatial data generated a positioning heatmap, see Figure 7, which provides an overview of the referee's "hot zones" during these sequences. Furthermore, to assess the system's transferability to another recording environment, three clips from a separate match (Match B) were processed. The extraction of movement trajectories in this second environment indicates that the calibration and tracking logic can be applied across different recording conditions.

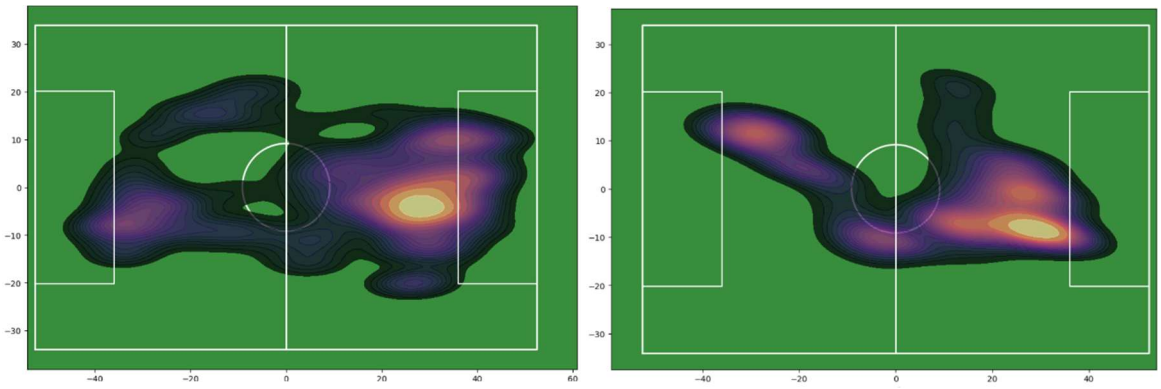


Figure 7: **Aggregated positioning heatmaps. Left: The primary dataset (Match A) generated from 14 clips. Right: The secondary validation dataset (Match B) generated from 3 clips, demonstrating cross-match reliability.**

4.3.2 Quantitative Performance Metrics

Beyond spatial mapping, the system extracted objective physical metrics from the processed sequences. For the primary dataset of 14 clips, the system calculated an accumulated total distance of 1027.7 meters, as detailed in Table 2. To illustrate the system's potential for targeted high-intensity analysis, a specific 5-second sprint interval was isolated from a single clip by manually defining its start and end timestamps. During this sequence, the system calculated a distance covered of 23.3 meters and a peak speed of 23.3 km/h, with an average speed of 16.7 km/h.

Table 2: **Extracted distance covered per video clip in the primary dataset.**

Clip	Distance Covered (meters)
1	51.2
2	62.9
3	91.9
4	67.5
5	69.9
6	81.7
7	69.3
8	70.4
9	90.4
10	77.7
11	92.2
12	84.2
13	36.4

14	82.1
Total	1027.7

Preliminary visual estimations of the video footage supported the plausibility of these calculated distances, indicating that the Proof of Concept provides a reasonable foundation for extracting movement metrics. These results suggest the feasibility of using accessible single-camera footage to generate objective performance metrics, providing a potential digital complement to the manual observation methods currently dominant at lower competitive levels.

4.4 Conceptual Application Design

The conceptual application prototype was developed in Figma to demonstrate how the extracted technical metrics and the identified domain needs could be translated into a practical self-analysis workflow. The prototype combines technically implemented outputs, such as positioning heatmaps, total distance, and sprint analysis, with conceptual features such as distance-to-ball visualization and event-based reflection tools. The purpose of this section is therefore to present the resulting application concept and show how the data pipeline can be connected to an accessible interface for referee self-analysis.

The prototype was structured around a match-review workflow in which the referee can select or upload a match, review video footage, inspect objective metrics, and add qualitative reflections. The primary objective was to demonstrate how a functional tool could be structured to facilitate objective self-analysis for football referees. The core performance metrics visualized in the design, such as positioning heatmaps, total distance, and sprint analysis, are based on the spatial data extracted during the technical phase of the project, as presented in Section 4.3. Additional contextual features, such as the distance-to-ball graph, were included as conceptual extensions based on the tracking capabilities of commercial systems such as Veo. Furthermore, the Figma prototype illustrates interactive features, such as timestamped notes and highlights, designed to support the qualitative aspects of self-reflection. These components are detailed below.

Upon logging in, the referee is presented with a dashboard displaying a history of their previously processed matches, which can be selected for detailed review. The interface also provides an intuitive workflow for uploading new match footage. Once a specific match is selected or uploaded, the user is directed to the main analysis view, which features a synchronized video player prominently at the top of the screen, see Figure 8.

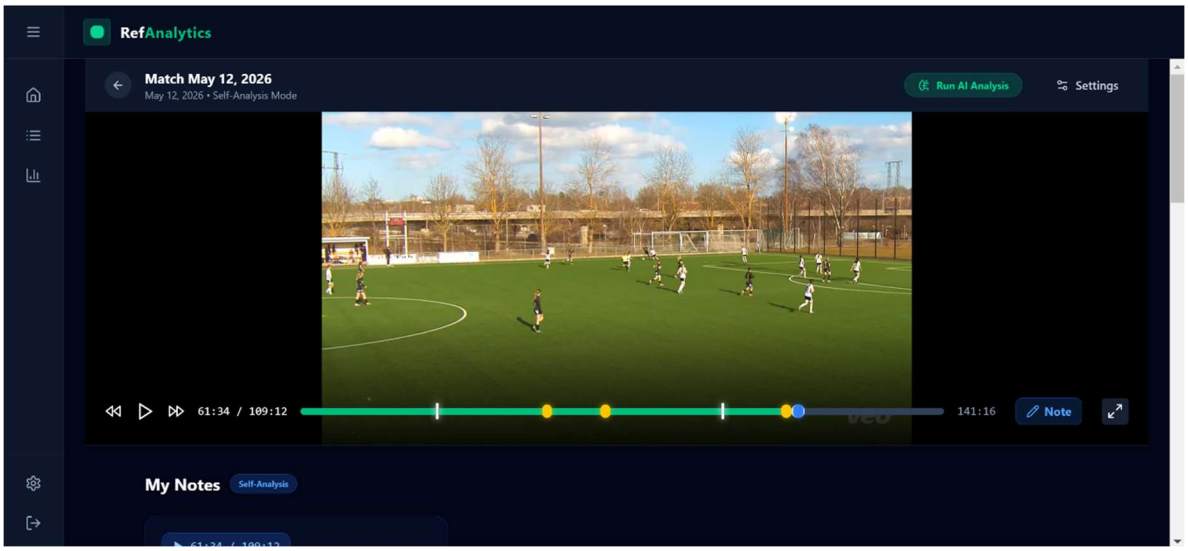


Figure 8: **The conceptual main analysis view in the RefAnalytics application, featuring a synchronized video player and an interactive timeline for match review.**

Below the video player, the user can calibrate the match by marking the exact start times of the first and second half. This synchronization allows the referee to efficiently review specific match events. For example, if a yellow card was issued in the 25th minute, the referee can jump directly to that timestamp to review the incident. This feature is particularly useful when cross-referencing an observer's report. If a match official has provided qualitative feedback with specific timestamps, the referee can immediately access those moments to better understand the observer's perspective.

The self-analysis process is further supported by the ability to create and manage highlights, as shown in Figure 9. These highlights are presented in a list, and the referee can review each situation by clicking the play button or by interacting with the color-coded markers on the timeline. Additionally, while reviewing the footage, the referee can add personal timestamped notes. These notes are automatically stored in a list and are visually represented as blue markers on the timeline, facilitating a direct link between subjective reflection and objective video evidence. In addition to event-specific feedback, the interface allows for general match comments that are not linked to a specific timestamp, providing a space for broader reflections on the overall performance.

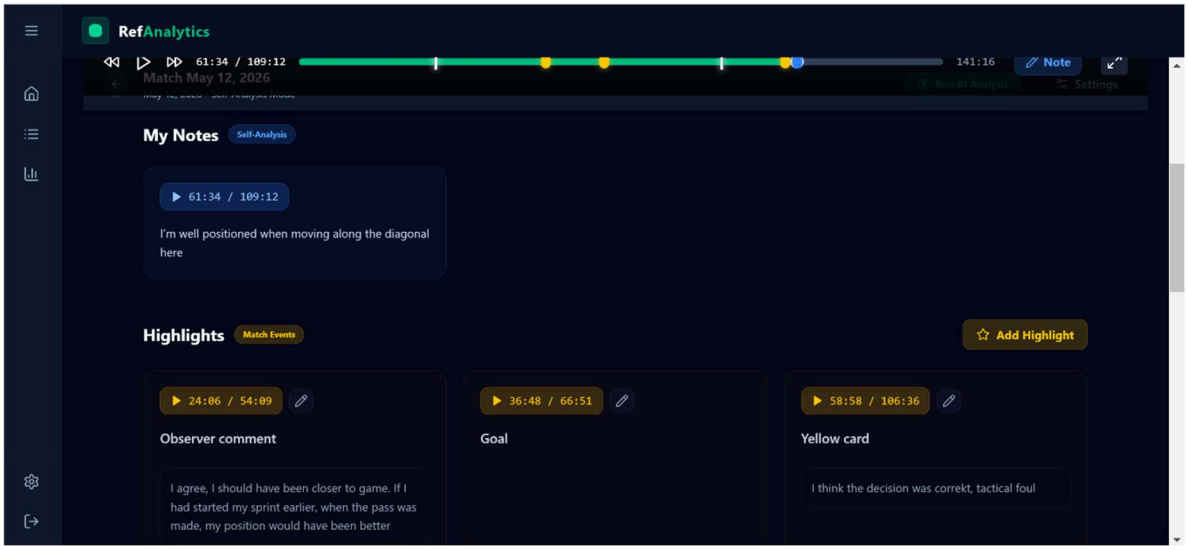


Figure 9: **The highlights and notes section of the interface, allowing referees to manage specific match events and add timestamped qualitative reflections.**

Another feature allows the referee to analyze specific high-intensity runs observed in the match footage. By manually entering the start and end timestamps of a sprint, the user can prompt the system to calculate the distance covered and peak speed for that specific interval. As demonstrated during the development of the technical data pipeline (see Section 4.2.3), the extraction of these physical metrics is functionally feasible. However, it should be noted that the absolute accuracy of these extracted sprint metrics has not yet been formally validated against ground-truth data (such as wearable GPS sensors). The conceptual layout for this sprint analysis module is presented in Figure 10.

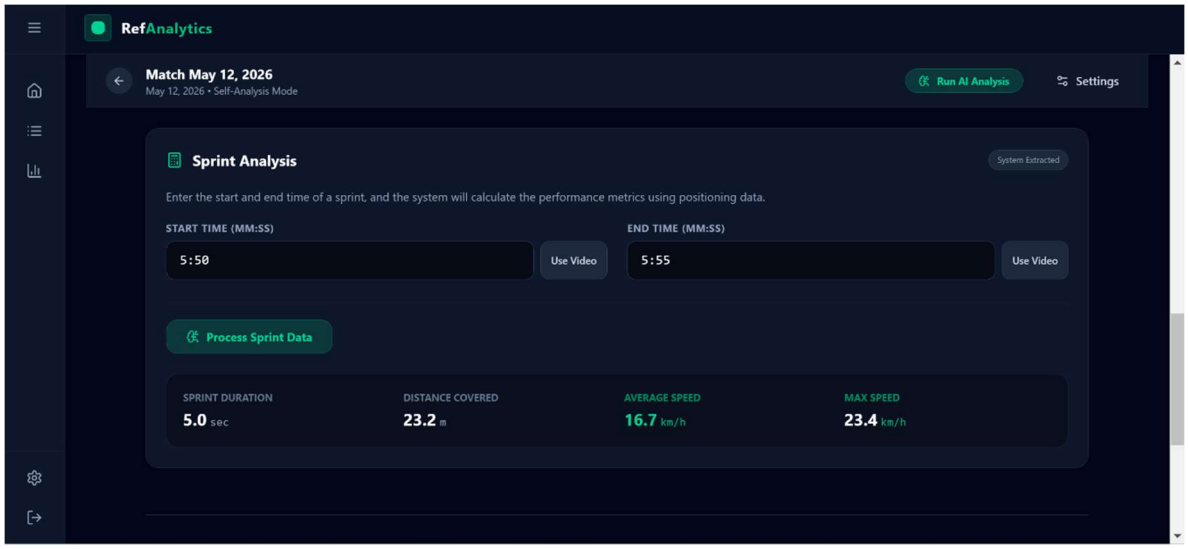


Figure 10: **The conceptual Sprint Analysis module, allowing referees to isolate specific timeframes to calculate targeted physical metrics such as distance covered and peak speed.**

Finally, the application processes the aggregate data to provide the referee with high-level performance insights. The primary dashboard, see Figure 11, features the Overall Positioning Heatmap. As demonstrated in the technical extraction phase, this visualization is generated directly

from the spatial data of the processed clips (e.g., the 14 clips from Match A). It provides the referee with an immediate, objective overview of their "hot zones," enabling them to visually assess whether they maintained the recommended diagonal positioning throughout the match. Additionally, the dashboard displays key physical metrics derived from the tracking engine, such as total distance covered. To further contextualize this physical effort, the interface concept includes a temporal graph illustrating the "Extracted Distance to Ball (Over Time)" see Figure 12. While "distance to ball" serves as a simplified proxy for "distance to active play," this continuous graph allows the referee to identify periods where they may have drifted too far from the action (e.g., during rapid counter-attacks), providing a specific focal point for further video review.



Figure 11: **The primary dashboard displaying the Overall Positioning Heatmap and high-level physical metrics, providing a comprehensive overview of the referee's match performance.**



Figure 12: **The temporal graph illustrating the referee's extracted distance to the ball over time, enabling targeted review of specific periods during the match.**

4.5 Qualitative Evaluation

To evaluate the practical utility of the conceptual application and the extracted data, a qualitative feedback session was conducted with Suhel Youssef, a referee developer at the Stockholm Football Association. The overall response was highly positive; the expert stated that the concept could become a valuable tool and noted its scalability, suggesting that it could be beneficial for referees at all competitive levels. Furthermore, the expert acknowledged the clear commercial potential of such a system.

4.5.1 Feature Validation and Usability

During the evaluation, Suhel Youssef noted that the manual features of the proposed application, such as timestamped highlights and notes, are reminiscent of the workflow currently used in *RefereeClips*, an internal match-review system used within the Swedish Football Association. However, the expert emphasized that the integration of automated objective metrics (such as heatmaps and sprint data) represents a significant advancement. By combining familiar manual review methods with the objective data pipeline, the proposed Proof of Concept provides a level of analysis that is currently missing even in professional internal systems.

Finally, the expert pointed out that although heatmaps, distance graphs and total distance covered are influenced by the specific characteristics of each match, they provide a valuable indication of performance. For example, if the data shows that a referee consistently fails to get closer than 20 meters to the play, it serves as a clear sign that the referee needs to develop their movement patterns to maintain better proximity to the action.

4.5.2 Identified Opportunities for System Expansion

When explicitly asked how the system could be further developed, the expert provided several ideas for expanding its functionality and analytical capabilities. To further streamline the workflow, the expert suggested enabling a direct integration between the official referee match reports and the application. Additionally, a specific interface improvement was recommended: integrating highlights directly into the "Distance to Ball" graph to provide an immediate visual correlation between the referee's physical positioning and key match events.

The expert also suggested utilizing the spatial coordinates for line-of-sight analysis to evaluate the referee's viewing angle and identify potential visual obstructions caused by players on the pitch. Furthermore, the expert highlighted the possibility of classifying movement types, where the existing tracking data could be used to distinguish between different locomotive patterns such as backward running or side-stepping. Another possibility discussed was extending the framework to include tracking of assistant referees, which would allow for a specialized evaluation of their side-stepping techniques and their ability to maintain precise alignment with the offside line during a match. Finally, the expert highlighted the value of adding spatial context to disciplinary incidents, for example by mapping the referee's position on the pitch when yellow or red cards were issued.

4.6 Post-Evaluation Design Refinements

Following the qualitative feedback session, an iterative design phase was conducted to incorporate the expert's suggestions into the conceptual application. This illustrates how the system can be continuously adapted to better serve the specific needs of referee developers.

To further streamline the workflow, the expert suggested enabling a direct integration between the official referee match reports and the application. Additionally, a specific interface improvement was recommended: integrating the event highlights directly into the "Distance to Ball" graph to provide an immediate visual correlation between the referee's physical positioning and key match events. Based on the recommendation to visually correlate physical positioning with match events (as discussed in Section 4.5.2), the "Distance to Ball" graph was redesigned. As shown in Figure 13, the application now integrates event highlights directly onto the temporal graph. To maintain a clean interface while maximizing utility, these highlight markers are interactive; hovering over a data point reveals a detailed tooltip displaying the specific event type, the timestamp, and the distance calculated at that moment. This interface update allows the referee to immediately identify if they were too far from the action when a crucial incident occurred.

Furthermore, building on the feedback regarding match-event integration, a new dedicated analytical view was developed. Figure 14 illustrates the "Card Incidents: Referee Positioning" mapping. By extracting the referee's coordinates at the timestamp when a card is issued, the system plots these events onto the 2D pitch. Similar to the temporal graph, these spatial markers are designed for interactivity, allowing the user to reveal the specific minute of the match by interacting with the plotted points. This enables officials to visualize their exact positioning when issuing disciplinary actions, helping them identify spatial patterns over time regarding where on the pitch critical decisions are made.



Figure 13: **Distance-to-ball graph with interactive event highlights integrated into the temporal visualization.**

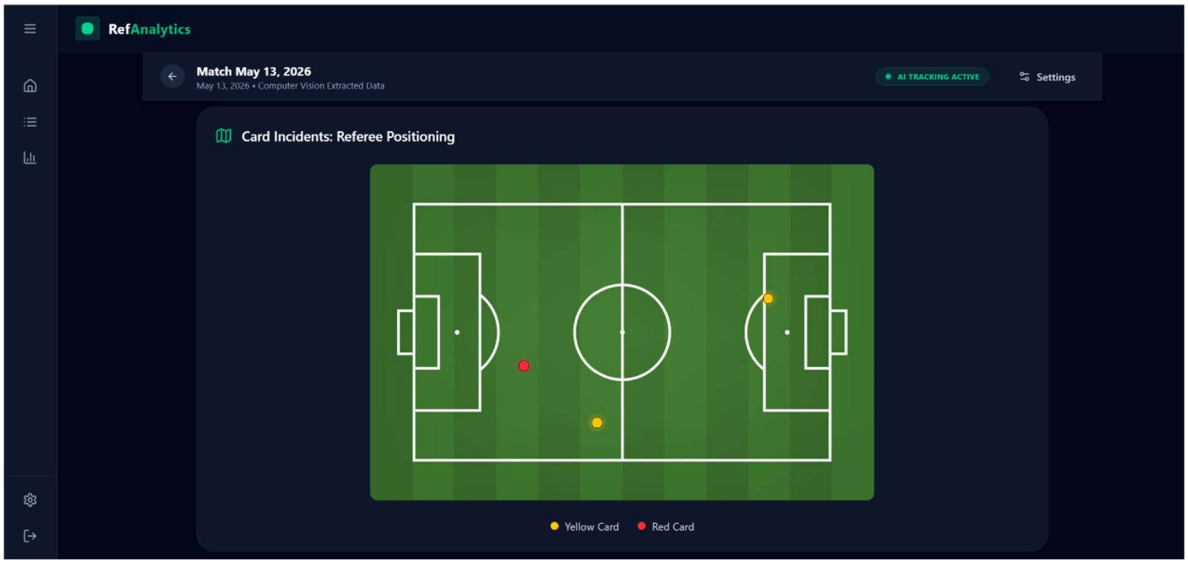


Figure 14: **Card incidents mapped onto the 2D pitch, showing the referee's positioning at the moment when disciplinary cards were issued.**

5 Analysis and Discussion

This section analyzes the results generated by the data pipeline and evaluates the practical applicability of the system based on the expert assessment. The discussion focuses on the reliability and validity of the extracted metrics, how the technical evidence relates to the conceptual application design, and the practical implications of the Proof of Concept. The section also connects the findings back to the three research questions, before addressing the main limitations of the study, outlining directions for future work, and reflecting on broader sustainability and ethical aspects of the proposed system.

5.1 Reliability Analysis

To evaluate the reliability of the extracted metrics, a manual plausibility assessment was conducted during the technical phase. Reliability in this context refers to the system's ability to consistently track movements and translate them into plausible physical distances. This assessment was performed in two steps. First, the referee's actual movement patterns in the video clips were visually compared against the plotted trajectories generated by the system, such as those shown in Figure 6. Second, running distances were estimated in 10-meter intervals to evaluate the total distance. Through both of these observational methods, a general correspondence with the system's calculated output was noted. However, it should be emphasized that this manual plausibility assessment does not constitute full technical validation, since it relies on human estimation rather than an independent ground-truth reference system. The evaluation should therefore be understood as a preliminary technical assessment of feasibility and plausibility, rather than a full validation of measurement accuracy. This aligns with the validation framework described in Section 2.5, where plausibility checks are presented as an initial indication rather than a substitute for formal validation against an independent reference system. To achieve higher scientific precision in future validation, wearable GNSS/GPS sensors or another validated EPTS reference system could have been used as an independent reference point (ground-truth) to statistically ensure the reliability of the metrics over time [15, 45].

5.2 Validity Analysis

The validity of the system, specifically, whether the system measures relevant parameters for referee development, was primarily evaluated through the expert evaluation with Suhel Youssef. This relevance is also supported by previous research identifying positioning-related variables, such as distance, angle, and insight, as important factors in referee performance and decision-making [6, 11]. These needs were also consistent with the author’s own domain experience as an active referee and referee coach, where positioning, proximity to play, and post-match reflection are recurring themes in referee development. In addition, video-based training and reflection have been shown to support football referees’ decision-making skills, which strengthens the relevance of combining spatial metrics with video-based self-analysis [22]. However, since the qualitative evaluation was based on feedback from one domain expert, the findings should be interpreted as indicative rather than generalizable. This also connects to the practical validation perspective discussed in Section 2.5.2, where the usefulness of a system depends not only on whether it can calculate metrics, but also on whether domain users can interpret and apply them [42]. The expert’s positive response suggests that the combination of objective metrics and qualitative video reflection may address a gap in current educational materials. Measuring the "distance to ball" can serve as a practical proxy for "distance to active play," which has been identified as an important factor for accurate decision-making. Furthermore, analyzing this distance over the course of a match can highlight if the gap generally increases over time, which may indicate fatigue and a need for improved physical endurance. During the interview, it was specifically highlighted that the manual components of the concept, such as notes and highlights, were perceived as aesthetically pleasing and significantly facilitate a referee’s ability to reflect on their own performance. Furthermore, the potential future integration of AI-generated feedback was viewed as an opportunity to create a unique and powerful tool.

Based on the expert's feedback, an iterative design process was conducted to further strengthen the analytical validity of the application in a practical context. A central update was the visualization of the referee's exact coordinates when issuing warnings and send-offs, information retrieved by the user manually entering the timestamp from the match report. This enables a direct discussion regarding whether the referee's positioning, for example, proximity to team officials, may have influenced the decision through external pressure. This type of connection between physical position and match events helps ensure that the tool does not only measure movement for its own sake but instead links the data to situations that are relevant for the referee's tactical development.

In the visualization of these disciplinary actions, the ball has been intentionally omitted from the pitch map. This is because warnings and send-offs frequently occur far from the active play, and the ball's location at these moments is therefore not a guaranteed indicator of the incident's context or the basis for the referee's decision. By focusing on the referee's position relative to, for example, team officials or specific zones rather than only the ball, the tool makes the data more relevant for the referee's tactical development.

5.3 Prototype Development Analysis

The development of the technical video analysis prototype showed that standard match footage can be transformed into objective referee movement data, but also that this process depends on several manual and semi-automatic steps. The video-based prototype was able to track the referee, transform pixel coordinates into pitch coordinates, and generate outputs such as running paths, heatmaps, distance covered, and sprint-related metrics. This indicates that the technical concept is feasible within the scope of a Proof of Concept. However, the development process also showed that the system is sensitive to tracking interruptions, camera movement, occlusion, and calibration accuracy. The video prototype should therefore be understood as a functional feasibility demonstration rather than a fully automated or validated tracking system.

The Figma prototype served a different purpose. While the video prototype tested whether objective spatial data could be extracted, the Figma prototype explored how such data could be presented in a practical self-analysis workflow. The development of the interface showed that raw metrics need to be connected to match video, notes, highlights, and event-based review in order to become meaningful for referee development. The Figma prototype therefore functioned as a bridge between the technical output and the intended user experience. It also made it possible to evaluate the concept with a domain expert before developing a full application.

Together, the two prototypes addressed different parts of the research questions. The technical video prototype demonstrated the feasibility of extracting referee performance data from standard match video, while the Figma prototype demonstrated how the extracted data could be translated into an accessible self-analysis concept. The combined prototype development therefore strengthened the overall feasibility assessment, but also highlighted that further work is needed to integrate the technical pipeline and the user interface into one automated and validated system.

5.4 Discussion

This section discusses the main technical and conceptual implications of the results. It focuses on the practical limitations encountered during tracking and clarifies the distinction between the components that were technically implemented in the Proof of Concept and the features that were conceptualized in the application design.

5.4.1 Handling tracking interruptions and occlusion

A technical challenge identified during the project was when the referee temporarily moved out of the camera's broadcast view or was obscured by players (occlusion). In the current data pipeline, these events cause the tracking algorithm to lose its target, requiring manual re-initialization.

To solve the issue of the referee moving out of frame, future versions of the system should apply the tracking algorithm directly to the camera's raw panoramic data rather than the cropped broadcast view. Optical systems like Veo use dual lenses to continuously record a 180-degree panoramic view of the entire pitch [47]. The standard view that follows the active play is merely a virtual zoom generated by software post-recording [47]. Since the underlying raw footage is intended to capture the full pitch, future AI-based tracking could potentially follow the referee more consistently regardless of the primary broadcast frame [47].

5.4.2 Technical evidence versus conceptual design

In line with the project delimitations, it is essential to distinguish between the technical deliverables and the visionary proposals presented in the high-fidelity Figma mockups. While the heatmaps and physical metrics have been generated through functional code and assessed through visual plausibility checks, a fully realized application has not been built, nor has an autonomous AI model been trained for this specific purpose. By synthesizing the empirically proven components with additional tracking capabilities, such as the ball data already provided by Veo, and presenting them within a user-friendly interface, the true potential of a future product is demonstrated.

A primary example of this is the calculation of the distance to the ball. Although the ball was not actively tracked within this Proof of Concept, the study demonstrates that such an integration is technically feasible. Since commercial camera systems such as Veo already provide spatial radar data including the ball and referee, combining the coordinate system developed in this project with such external tracking data can be understood as a feasible extension rather than a separate technical barrier [8].

5.4.3 Practical implications and limitations

While the technical results and expert feedback indicate that the concept has practical potential, the current Proof of Concept should be understood as an early-stage feasibility demonstration rather than a finished evaluation tool. Nevertheless, the results demonstrate a successful Proof of Concept, as the system was able to extract referee movement data from standard match footage, generate spatial visualizations and physical metrics, and translate these outputs into a practically relevant application concept. The semi-automatic workflow was suitable for investigating whether referee movement data could be extracted from standard match footage, but a practical end-user application would require a higher degree of automation. In addition, the qualitative evaluation was based on one domain expert, which provided valuable professional insight but limits the generalizability of the findings. Future evaluations should therefore include a larger group of referees and referee developers to assess whether the interface, metrics, and workflow are useful across different experience levels and match contexts.

From an environmental perspective, the concept may limit the need for additional physical tracking hardware by reusing already available match recordings and camera infrastructure, which is relevant since electronic devices can contribute to e-waste if they are not managed and recycled properly [9, 10, 48]. From an economic perspective, the use of existing single-camera match footage may reduce dependence on more advanced tracking setups and make structured referee development more accessible at non-elite levels [10, 46]. From a social perspective, more accessible feedback tools could contribute to more equal development opportunities for referees outside professional environments and support fair, inclusive, and well-functioning sport environments [17, 19].

5.5 Research Questions

This section summarizes how the results answer the three research questions introduced in Section 1.2. The questions correspond to the three main parts of the study: understanding the domain need, testing technical feasibility, and evaluating how the extracted data can be presented in a practical self-analysis concept.

RQ1 – Domain understanding: The study found that valuable referee performance analysis should combine objective spatial information with the practical situations that referees and referee coaches already use when reviewing performance. Based on literature, domain knowledge, and expert feedback, the most relevant indicators were positioning patterns, distance to active play, distance covered, sprint intervals, event-based positioning, and the possibility to connect these metrics to video clips, highlights, and written reflections. This indicates that referee performance analysis should not only focus on isolated numerical values, but also on how these values can support interpretation, coaching, and self-reflection.

RQ2 – Technical feasibility: The Proof of Concept demonstrated that objective referee performance metrics can be extracted from standard football match recordings to a limited but meaningful extent. By combining object tracking, homography calibration, coordinate projection, and post-processing, the system generated referee trajectories, heatmaps, total distance values, and sprint-related metrics from selected video clips. However, the results should be interpreted as a feasibility demonstration rather than a fully validated measurement system, since the extracted data has not yet been validated against an independent ground-truth reference system.

RQ3 – Design and evaluation: The extracted metrics can be presented in a self-analysis application by combining quantitative outputs with match video, event highlights, notes, and spatial visualizations. In this context, the manual elements are an intentional part of the self-analysis workflow, since they allow referees to connect objective data with their own reflections and key match situations. The Figma prototype showed how heatmaps, distance graphs, sprint metrics, and event-based referee positioning can be organized into a workflow where referees review match situations and reflect on their own performance. The expert feedback indicated that the concept has practical potential, particularly as a tool for supporting referee development and structured post-match reflection.

5.6 Limitations

The primary limitation of this study was the restricted timeframe, which impacted the scope of the user testing. Although an in-depth qualitative interview was conducted with a referee developer, there was insufficient time to include a broader user study involving a larger group of active referees to evaluate the tool's daily usability in the field. An additional consequence of the limited time was the lack of opportunity to further investigate and attempt to implement more of the specific suggestions provided during the expert interview.

5.7 Future work

To move beyond this technical Proof of Concept, several future development steps are recommended. A natural next step is to actually build the application visualized in the study's Figma prototypes to make the technology accessible to end-users. Within this development, efforts should be made to implement the remaining suggestions identified during the expert evaluation to further enhance the tool's practical value. To enable large-scale use, the system should be fully automated by implementing object detection models (e.g., YOLO), which would eliminate the need for manual initialization. Furthermore, integrating AI-generated feedback, line-of-sight analysis, and tracking of assistant referees would provide even deeper tactical insights. Future studies should also validate

the extracted spatial metrics against an independent reference system, such as GNSS/GPS or another validated EPTS technology, to assess the accuracy of the distance and speed calculations more rigorously One possible validation approach would be to equip the referee with a validated GPS/GNSS vest during a Veo-recorded match, allowing the coordinates generated by the optical system to be synchronized and compared against the sensor-based reference data.

5.8 Reflections

From a sustainability perspective, the proposed system may contribute to social sustainability by making structured performance feedback more accessible to referees outside elite-level environments, which connects to the broader role of sport in supporting inclusion and sustainable development, while also requiring careful ethical handling of personal tracking data [18]. From an environmental perspective, reusing already recorded match footage may also reduce the need for additional dedicated hardware or travel for in-person observation, although the energy use of cloud-based video processing should be considered in future implementations. From a social and economic perspective, the work contributes to democratizing access to performance data. Historically, advanced sports analytics have primarily been reserved for elite teams with substantial financial resources [3]. By demonstrating that a single-camera setup can generate valuable spatial data, this project creates opportunities for grassroots and semi-professional referees to engage in reflective practice and improve their skills [4, 22].

From an ethical standpoint, increased tracking and surveillance of individuals raises questions regarding personal privacy. It is of utmost importance that referees do not perceive the system as a form of constant surveillance; the system must exist for their benefit and act as a support in their development. If implemented broadly, the system should primarily be utilized as an internal educational tool owned by the referees themselves, rather than a punitive surveillance system controlled by external parties. This does not preclude observers and referee coaches from benefiting from the system to enhance the guidance they already provide on-site. Provided the use is benevolent and focused on personal growth, the technology can serve as a shared foundation to deepen the coaching dialogue. Insights gathered during the study indicate that the technology, if applied correctly, has the potential to support referee development across several competitive levels. In this way, technology becomes accessible to all referees, regardless of level, and contributes to a more objective and fair evaluation culture.

6 Conclusion

The primary purpose of the report was to address the lack of objective tools for performance evaluation of football referees at a non-elite level. By developing a Proof of Concept for a video analysis data pipeline and conceptualizing an application for self-analysis, the project has shown that it is technically feasible to extract relevant spatial data from standardized match recordings. The work has thus contributed to bridging the technical gap between player analysis and referee evaluation by demonstrating how existing infrastructure for video recording can be adapted for match officials.

The research questions can therefore be answered affirmatively within the scope of a Proof of Concept: objective referee positioning data can be extracted from standard match video, and the extracted data can be used as a basis for designing a meaningful self-analysis application. However, the technical accuracy of the extracted metrics requires further validation before the system can be considered fully reliable for practical deployment.

The system's ability to visualize the referee's general movement patterns through heatmaps and calculate distances to the active play constitutes central results. These visualizations enable the referee to identify crucial tendencies in their positioning, such as the application of a diagonal movement pattern. Since correct positioning is an important factor for accurate decision-making, this provides the referee with concrete insights for their development [6].

Furthermore, the evaluation with Suhel Youssef confirmed that the system is both relevant and possesses high practical value for referee development at all levels. The expert’s positive response suggests that the combination of objective metrics and qualitative video reflection may address a gap in current educational materials. The expert evaluation suggested that such a video-based tool for reflective practice has potential to support a referee's self-awareness and tactical understanding.

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